

4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral $\chi^2(1)$ pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on $\mathbf{A}$ and Σ , and shows that for the spherical case the condition is necessary as well as sufficient.

4.3.1 Spherical normal vectors

Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$, let $\mathbf{A}$ be symmetric, and let $r \geq 1$ be an integer and $\gamma \geq 0$. Then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ if and only if $\mathbf{A}$ is idempotent with $\text{rank}(\mathbf{A}) = r$ and $\gamma = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

Proof. Sufficiency. By Theorem 1.9.2, $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^T$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns and $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$. Then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = \|\mathbf{Q}_1^T \mathbf{Y}\|^2$ and $\mathbf{Q}_1^T \mathbf{Y} \sim N_r(\mathbf{Q}_1^T \boldsymbol{\mu}, \mathbf{Q}_1^T \mathbf{I}_n \mathbf{Q}_1) = N_r(\mathbf{Q}_1^T \boldsymbol{\mu}, \mathbf{I}_r)$, so by definition $\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^T \boldsymbol{\mu}\|^2)$, and $\|\mathbf{Q}_1^T \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

Necessity. Apply Theorem 4.2.1 with $\mathbf{L} = \mathbf{I}$, so that $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^T$. Let $\boldsymbol{\nu} = \mathbf{Q}^T \boldsymbol{\mu}$. Then $\mathbf{b} = \mathbf{Q}^T \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \boldsymbol{\nu}$, so $b_j = \lambda_j \nu_j$. The constant left after completing squares is $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \boldsymbol{\nu}^T \boldsymbol{\Lambda} \boldsymbol{\nu} - \sum_j \lambda_j \nu_j^2 = 0$. Hence $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has the law of $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$, where J indexes the nonzero eigenvalues. J is not empty, because a $\chi^2(r, \gamma)$ variable with $r \geq 1$ is not identically zero.

By Lemma 4.1.1, near $t = 0$ the logarithm of the mgf of this sum is $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$. Expanding $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$ and $x / (1 - x) = \sum_{k \geq 1} x^k$, the coefficient of t^k is $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k \nu_j^2)$. The same expansion of Eq. 4.1.2 gives $(2^{k-1}/k)(r + k\gamma)$. Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k \nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let $\rho = \max_{j \in J} |\lambda_j|$. For even k every term on the left is nonnegative and one of them is at least ρ^k . The right side grows only linearly in k , so $\rho \leq 1$. Split J into $J_+ = \{\lambda_j = 1\}$, $J_- = \{\lambda_j = -1\}$ and $J_0 = \{0 < |\lambda_j| < 1\}$, and write $N_\pm = |J_\pm|$ and $S_\pm = \sum_{j \in J_\pm} \nu_j^2$. The terms over J_0 tend to zero as $k \rightarrow \infty$, so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by k and letting $k \rightarrow \infty$ through even and through odd values gives $S_+ + S_- = \gamma = S_+ - S_-$. Then the same limit without the factor k gives $N_+ + N_- = r = N_+ - N_-$. So $N_- = S_- = 0$, $N_+ = r$ and $S_+ = \gamma$. The terms over J_+ in Eq. 4.3.1 now account for all of $r + k\gamma$, and with $k = 2$ the remaining sum $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$ is zero. Every

term of that sum is positive, so J_0 is empty. Thus every eigenvalue of $\mathbf{A}$ is 0 or 1, $\mathbf{A}$ is idempotent, $\text{rank}(\mathbf{A}) = N_+ = r$, and $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$. ■

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

4.3.2 General covariance matrices

Theorem 4.3.2 — When a quadratic form is chi-squared

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric.

- (a) (*Projections.*) If $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ with $\sigma > 0$ and $\mathbf{P}$ is symmetric and idempotent, then

$$\frac{\|\mathbf{P}\mathbf{Y}\|^2}{\sigma^2} = \frac{\mathbf{Y}^T \mathbf{P} \mathbf{Y}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite $\boldsymbol{\Sigma}$.*) If $\boldsymbol{\Sigma}$ is positive definite, then $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has a $\chi^2(r, \gamma)$ distribution for some $r \geq 1$ and $\gamma \geq 0$ iff $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\mathbf{A} \neq \mathbf{0}$. In that case $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ and $\gamma = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

- (c) (*Any $\boldsymbol{\Sigma}$.*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu},$$

then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu})$. The three conditions hold in particular when $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$, and when $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$.

Proof.

- (a) $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Apply Theorem 4.3.1 with $\mathbf{A} = \mathbf{P}$, using $\mathbf{P}^T \mathbf{P} = \mathbf{P}$ to write $\mathbf{Y}^T \mathbf{P} \mathbf{Y} = \|\mathbf{P}\mathbf{Y}\|^2$ and $\boldsymbol{\mu}^T \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$.
- (b) Let $\boldsymbol{\Sigma}^{1/2}$ be the positive definite square root (Theorem 1.7.6). Then $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$ and $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = \mathbf{Z}^T \mathbf{B} \mathbf{Z}$ with $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ symmetric. By Theorem 4.3.1, $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ is noncentral chi-squared iff $\mathbf{B}$ is idempotent and nonzero. Now $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$, so $\mathbf{B}^2 = \mathbf{B}$ iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ (multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides) iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$ (multiply by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$ on the right). Also $\mathbf{B} \neq \mathbf{0}$ iff $\mathbf{A} \neq \mathbf{0}$. The degrees of freedom are $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$, which equals $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$, and the noncentrality is $\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.
- (c) Take $\mathbf{L}$ of size $n \times k$ with full column rank $k = \text{rank}(\boldsymbol{\Sigma})$ and $\mathbf{L} \mathbf{L}^T = \boldsymbol{\Sigma}$, for instance $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$ from the eigenvectors $\mathbf{U}_1$ for the positive eigenvalues $\mathbf{D}_1$ of $\boldsymbol{\Sigma}$. (If $\boldsymbol{\Sigma} = \mathbf{0}$ the claim is trivial.) As in Theorem 4.2.1, $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$ with $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. Since $\mathbf{L}$ has full column rank, $\mathbf{L}^T$ has full row rank, so $\mathbf{L} \mathbf{G} \mathbf{L}^T = \mathbf{L} \mathbf{H} \mathbf{L}^T$ implies $\mathbf{G} = \mathbf{H}$, and $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$ implies $\mathbf{g} = \mathbf{h}$. Put $\mathbf{B} = \mathbf{L}^T \mathbf{A} \mathbf{L}$ and $\mathbf{c} = \mathbf{L}^T \mathbf{A} \boldsymbol{\mu}$. Condition (i) reads $\mathbf{L} \mathbf{B}^2 \mathbf{L}^T = \mathbf{L} \mathbf{B} \mathbf{L}^T$, so $\mathbf{B}^2 = \mathbf{B}$. Condition (ii) reads $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$, so

$\mathbf{B}\mathbf{c} = \mathbf{c}$. Condition (iii) reads $\mathbf{c}\mathbf{T}\mathbf{c} = \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$. Hence

$$(\boldsymbol{\mu} + \mathbf{L}\mathbb{Z})\mathbf{T}\mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbb{Z}) = \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu} + 2\mathbf{c}\mathbf{T}\mathbb{Z} + \mathbb{Z}\mathbf{T}\mathbf{B}\mathbb{Z} = (\mathbb{Z} + \mathbf{c})\mathbf{T}\mathbf{B}(\mathbb{Z} + \mathbf{c}),$$

because $(\mathbb{Z} + \mathbf{c})\mathbf{T}\mathbf{B}(\mathbb{Z} + \mathbf{c}) = \mathbb{Z}\mathbf{T}\mathbf{B}\mathbb{Z} + 2\mathbf{c}\mathbf{T}\mathbf{B}\mathbb{Z} + \mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c}$ and $\mathbf{B}\mathbf{c} = \mathbf{c}$. Now $\mathbb{Z} + \mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$, and [Theorem 4.3.1](#) gives the law $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c})$, where $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ and $\mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c} = \mathbf{c}\mathbf{T}\mathbf{c} = \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$. If $\mathbf{B} = \mathbf{0}$, then $\mathbf{c} = \mathbf{B}\mathbf{c} = \mathbf{0}$ and the form is identically 0, which is $\chi^2(0)$. If $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A}$, all three conditions follow by substitution. If $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\boldsymbol{\mu} = \boldsymbol{\Sigma}\mathbf{v}$, then (i) follows from multiplying $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma} = \mathbf{A}\boldsymbol{\Sigma}$ by $\boldsymbol{\Sigma}$ on the left, and (ii) and (iii) from the same identity applied to $\mathbf{v}$. ■

Part (a) is the case used most often in this book. For the linear model $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and the projection $\mathbf{M}$ onto $\mathbb{C}(\mathbf{X})$, [Chapter 6](#) shows that the residual sum of squares is $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$. Since $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of $\boldsymbol{\beta}$. The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of $\mathbf{A}\boldsymbol{\Sigma}$ alone is *not* enough when $\boldsymbol{\Sigma}$ is singular. A mean with a component outside $\mathbb{C}(\boldsymbol{\Sigma})$ is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

Corollary 4.3.3 — Mahalanobis forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite, $(\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$ and $\mathbf{Y}\mathbf{T}\boldsymbol{\Sigma}^{-1}\mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}\mathbf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})$.
- (b) If $\mathbf{G}$ is a symmetric generalized inverse of $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$, then $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y} \sim \chi^2(\text{rank}(\boldsymbol{\Sigma}), \boldsymbol{\mu}\mathbf{T}\mathbf{G}\boldsymbol{\mu})$.

Proof.

- (a) $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ satisfies $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A}$, and $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$. The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b) $\mathbf{G}\boldsymbol{\Sigma}$ is idempotent because $\boldsymbol{\Sigma}\mathbf{G}\boldsymbol{\Sigma} = \boldsymbol{\Sigma}$, so part (c) of the theorem applies and the degrees of freedom are $\text{tr}(\mathbf{G}\boldsymbol{\Sigma}) = \text{tr}(\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L})$. With $\mathbf{L}$ as in that proof, $\mathbf{L}\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L}\mathbf{L}\mathbf{T} = \mathbf{L}\mathbf{L}\mathbf{T}$ gives $\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L} = \mathbf{I}_k$, whose trace is $k = \text{rank}(\boldsymbol{\Sigma})$. ■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y}$ does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

Example. (Correlated observations about a common level)

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite, and suppose we want to measure how far $\mathbf{Y}$ is from a common level $m\mathbf{1}$. The generalized least squares level is $\hat{m} = \mathbf{1T}\boldsymbol{\Sigma}^{-1}\mathbf{Y}/\mathbf{1T}\boldsymbol{\Sigma}^{-1}\mathbf{1}$, and the natural discrepancy is $(\mathbf{Y} - \hat{m}\mathbf{1})\mathbf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{YTA}\mathbf{Y}$ with

$$\mathbf{A} = \boldsymbol{\Sigma}^{-1} - \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}\mathbf{1T}\boldsymbol{\Sigma}^{-1}}{\mathbf{1T}\boldsymbol{\Sigma}^{-1}\mathbf{1}}.$$

A direct multiplication gives $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A}$, and $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = n - 1$. By [Theorem 4.3.2\(b\)](#) the form is noncentral chi-squared with $n - 1$ degrees of freedom and noncentrality $\boldsymbol{\muT}\mathbf{A}\boldsymbol{\mu}$, which is zero exactly when $\boldsymbol{\mu}$ is constant. The listing takes $n = 8$, an autoregressive correlation $\boldsymbol{\Sigma} = (\rho^{|i-j|})$ with $\rho = 0.6$, and a linear trend in the mean, for which $\gamma = 4.200$. [Figure 4.3.1\(a\)](#) compares 200,000 simulated values with the $\chi^2(7, 4.200)$ density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

[Run this code online](#)

4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean, $\mathbf{YTCY}$ with $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1T}$, is an example as soon as the observations are correlated. Then $\mathbf{C}\boldsymbol{\Sigma}$ is not idempotent, and by [Theorem 4.2.1](#) the law is that of $\sum_j \lambda_j (W_j + \nu_j)^2$, with weights λ_j equal to the nonzero eigenvalues of $\mathbf{C}\boldsymbol{\Sigma}$. In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

Exact computation. The characteristic function of the weighted sum is known in closed form (put $t = iu$ in [Corollary 4.2.2](#)), and a distribution function can be recovered from a characteristic function φ by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\text{Im}\{e^{-iux}\varphi(u)\}}{u} du.$$

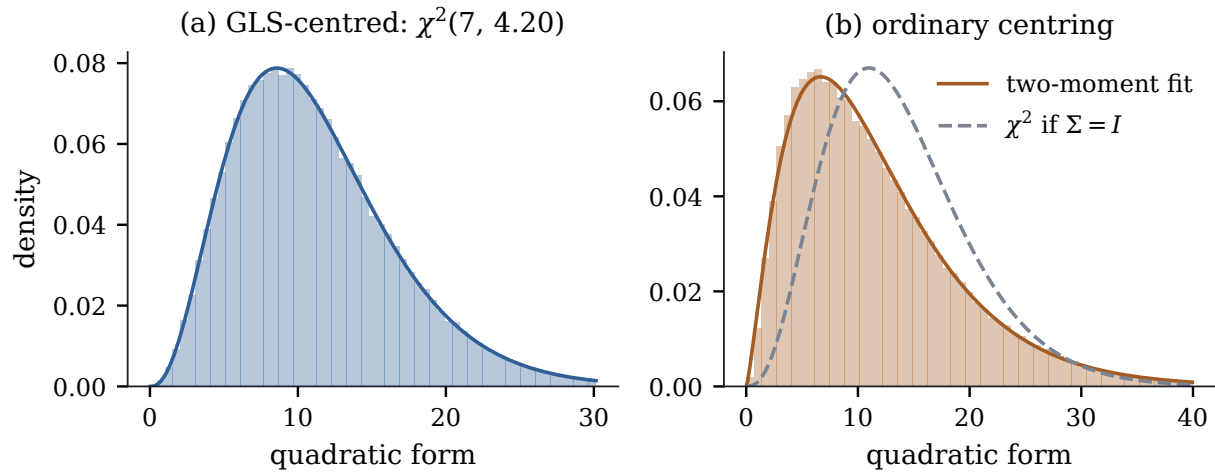


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for $n = 8$ autoregressive observations with a trending mean. (a) With generalized least squares centring, $\mathbf{A}\Sigma$ is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral $\chi^2(1)$ terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

For a weighted sum of k chi-squared terms the integrand decays like $u^{-1-k/2}$, so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

Two-moment approximation. Match $a\chi^2(\nu)$ to the mean m and variance v from Theorem 4.2.3. Since $a\chi^2(\nu)$ has mean $a\nu$ and variance $2a^2\nu$, this gives $a = v/(2m)$ and $\nu = 2m^2/v$, with ν usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of **YTCY** is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with $a = 2.488$ and $\nu = 4.68$ gives 0.0484. Treating the observations as independent, and so using $\chi^2(7, 6.720)$, gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of ρ , but strong correlation makes it severe: with $\rho = 0.9$ and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)"""
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
```

```

 $\gamma(-1-k/2)$ 
val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges
[1:]))
return 0.5 + val / np.pi

```

[Run this code online](#)

4.3.4 Exercises

4.3.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ with $n \geq 2$, and $\mathbf{A} = \alpha \mathbf{I} + \beta \mathbf{1}\mathbf{1}^T$. Find all (α, β) for which $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, and identify each form.

Solution. $\mathbf{A}$ has eigenvalue $\alpha + n\beta$ on $\mathbf{1}$ and α on $\mathbf{1}^\perp$. By [Theorem 4.3.1](#), the form is chi-squared iff $\mathbf{A} \neq \mathbf{0}$ and both eigenvalues lie in $\{0, 1\}$. The three solutions are $(1, 0)$, giving $\|\mathbf{Y}\|^2 \sim \chi^2(n)$; $(0, 1/n)$, giving $n\bar{Y}^2 \sim \chi^2(1)$; and $(1, -1/n)$, giving $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$.

Exercise A2.

Two independent samples of sizes m and k come from $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$. Write the pooled within-sample sum of squares as $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ for a suitable projection, and show that the two-sample t statistic has the $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m+1/k}))$ distribution.

4.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T\})$ with $-1/(n-1) < \rho < 1$, and let $\bar{Y}$ and S^2 be the sample mean and variance. Show that $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$ and that S^2 is independent of $\bar{Y}$. Deduce that $\sqrt{n}(\bar{Y} - \mu)/S$ has the law of $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$ times a $t(n-1)$ variable, and describe what a positive ρ does to the coverage of the usual 95% t interval for μ .

Exercise B2.

Let $\Sigma = \text{diag}(1, 0)$, $\mu = (0, 1)^T$, $\mathbf{A} = \mathbf{I}_2$ and $\mathbf{Y} \sim N_2(\mu, \Sigma)$. Show that $\mathbf{A}\Sigma$ is idempotent but $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if $\mu = (m, 0)^T$? Repeat the first part with $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, where the shift enters through a linear term.

Solution. $\mathbf{A}\Sigma = \Sigma$ is idempotent. But $Y_2 = 1$ with probability one, so $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$, while every $\chi^2(r, \gamma)$ with $r \geq 1$ gives positive probability to $(0, 1)$. Conditions (i) and (ii) hold, since $\Sigma^3 = \Sigma$ and $\Sigma^2\mu = \mathbf{0} = \Sigma\mu$. Condition (iii) fails, since $\mu^T \Sigma \mu =$

$0 \neq 1 = \boldsymbol{\mu}^T \boldsymbol{\mu}$. The part of the mean outside $\mathbb{C}(\boldsymbol{\Sigma})$ adds a constant. If $\boldsymbol{\mu} = (m, 0)^T \in \mathbb{C}(\boldsymbol{\Sigma})$, then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$, as [Theorem 4.3.2\(c\)](#) predicts. For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $\boldsymbol{\mu} = (0, 1)^T$, $\mathbf{A}\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ is again idempotent, and $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$, which is negative with positive probability. Conditions (i) and (ii) hold, since $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma}$ and $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = (1, 0)^T$, but (iii) fails because $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = 1 \neq 0 = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$. The shift now enters through the linear term $2Y_1$.

Exercise B3.

In [Corollary 4.3.3\(b\)](#), show that $\mathbf{Y}^T \mathbf{G} \mathbf{Y}$ takes the same value for every generalized inverse $\mathbf{G}$ of $\boldsymbol{\Sigma}$, with probability one. *Hint:* Theorem 2.2.3.

Solution. By Theorem 2.2.3, $\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$ with probability one. Since $\boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$, also $\mathbf{Y} \in \mathbb{C}(\boldsymbol{\Sigma})$, so $\mathbf{Y} = \boldsymbol{\Sigma} \mathbf{v}$ for some (random) $\mathbf{v}$. For any generalized inverse $\mathbf{G}$, $\mathbf{Y}^T \mathbf{G} \mathbf{Y} = \mathbf{v}^T \boldsymbol{\Sigma} \mathbf{G} \boldsymbol{\Sigma} \mathbf{v} = \mathbf{v}^T \boldsymbol{\Sigma} \mathbf{v}$, which does not involve $\mathbf{G}$.

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite, and let $\mathbf{A} \neq \mathbf{0}$ be nonnegative definite. Show that the two-moment approximation $a\chi^2(\nu)$ to $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has $\nu = \{\text{tr}(\mathbf{A}\boldsymbol{\Sigma})\}^2 / \text{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2\}$, and that $\nu \leq \text{rank}(\mathbf{A})$, with equality iff all nonzero eigenvalues of $\mathbf{A}\boldsymbol{\Sigma}$ are equal.

4.3.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ possibly singular, and suppose $\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ with $r \geq 1$. Show that $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}$. *Hint:* in [Corollary 4.2.2](#), compare the coefficients of t^k for $k \geq 3$ with those of [Eq. 4.1.2](#), and adapt the argument of [Theorem 4.3.1](#).